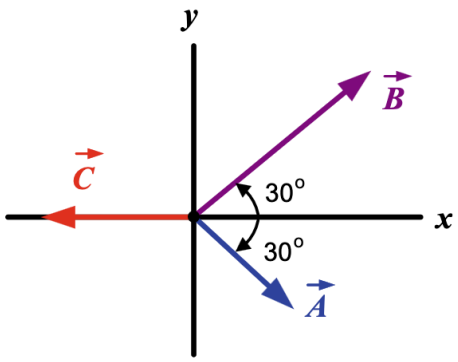
Worksheet 1

Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).

Problems

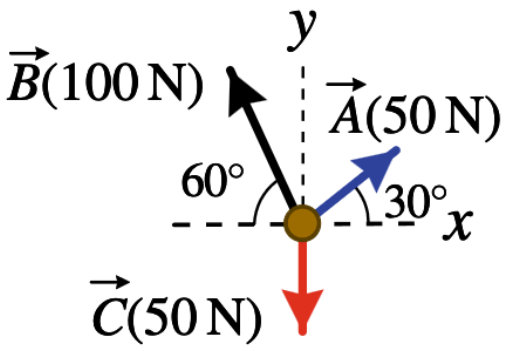
This is a worksheet to practice working with vectors.



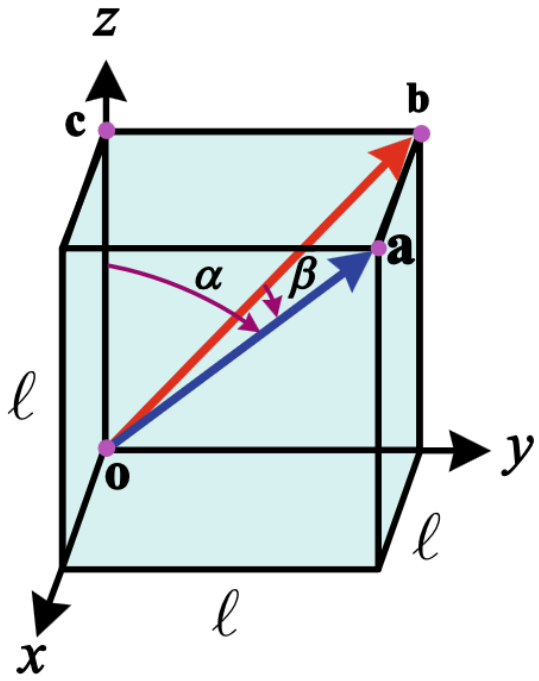
1. Three vectors are oriented as shown in the image above, where $A = 10$, $B = 20$, and $C = 15$ units. Find:

   (a) the x and y components of the resultant vector $\vec{D} = \vec{A} + \vec{B} + \vec{C}$,

   (b) The magnitude and direction of the resultant vector.



2. Three horizontal forces represented by the vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$, are acting on a point as shown in the figure above. Find the magnitude and direction of a fourth force on the point that would make the vector sum of the four forces equal to zero.



3. The figure above shows a cube of side ℓ with one corner being the origin of the Cartesian coordinate system

   (a) Use vectors to find the angle α between the z-axis and the diagonal from the origin to th opposite corner at point a.

   (b) Find the angle β between the line $\overline{oa}$ and the line $\overline{ob}$.